

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester Bachelor of Technology (B.Tech) (CE/IT)

University Examination November 2010

CE 201 Data Structures & Algorithms

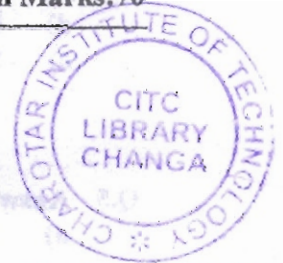
Date: 26.11.2010, Friday

Time: 10:30 a.m to 1:30 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.



SECTION – I

Q.1 Answer the following questions.

- a) What is linear and non linear data structure? Explain with a suitable example. 02
- b) What is recursion? Write a recursive algorithm to find out the factorial of a given number. 02
- c) What are the advantages of converting an arithmetic expression from infix notation to postfix notation? 02
- d) What are the differences between array & linked list? 03
- e) State True or False with justification:
 - (i) Sparse array has high-density of non-zero elements.
 - (ii) All primitive recursive functions can be solved iteratively.
 - (iii) Stack works in First in last out and Queue works in Last in First out.

FOR REFERENCE

Q.2 Answer the following questions.

- a) Show circular queue contents with front and rear after each step with size=5. 04
 - (i) Insert 10, 20, 30. (ii) Delete (iii) Insert 40, 50, 60, 70. Initially queue is empty.
- b) What is hashing? Explain any three hashing techniques with example. 04
- c) Convert the following infix expression into its equivalent postfix expression using stack. $(A+B) * D + E / (F + A * D * E) + C$ 04

OR

Q.2 Answer the following questions

- a) Given a two dimensional array Z1 (3:10, 10:20) stored in row-major order with base address of 200 and size of each element of 4 bytes, find address of element Z1(5, 15). 04
- b) Write a short note on indexed sequential file organization. 04
- c) Define double ended queue. Explain the operations on it with suitable example. 04

Q.3

- a) Write an algorithm to solve Tower of Hanoi problem for n=3. Trace the algorithm using stack. 06
- b) Generate index for each data using hash function and arrange them into an array A[10]. Use linear probing to resolve collision. 06
Data: 1104, 806, 1288, 697, 1087, 835, 533, 1407
Hash function $H(x) = (x \bmod 23) + 2$.

OR

Q.3 Answer the following questions.

- a) Write and explain diagrammatically the algorithm for insertion of a node in doubly linked list. 06
- b) Write an algorithm to copy one stack to another stack assuming the stack is implemented using array. 06

SECTION – II

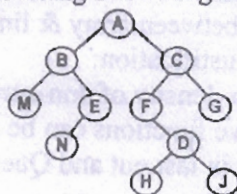
Q.4

Answer the following questions.

- a) Define the terms: (i) Graph (ii) Path Matrix 02
- b) Differentiate: Linear search - Binary search 02
- c) Explain simple merge sort with suitable example. 02
- d) Explain fields and records in a file. 02
- e) State True or False with justification: 03
 - (i) Inorder-traversal of a binary search tree arranges data into ascending order.
 - (ii) Depth first search technique can only be used for undirected graph.
 - (iii) The indegree of a vertex is the number of edges leaving it.

Q.5 Answer the following questions.

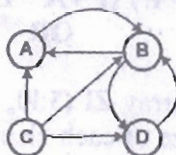
- a) Sort the following data using Radix sort. Trace the algorithm. 04
103, 12, 150, 1405, 145, 1450, 130, 1045
- b) Construct the Binary Search Tree using following data. Show each step. 04
32, 45, 12, 11, 13, 92, 78, 66, 17, 70, 98, 108.
Delete 92 and then 32 from the tree. Draw tree after each deletion.
- c) Assign the threads to the NULL links of given binary tree for inorder traversal. 04
Redraw the tree with head node using following structure of a node.



OR

Q.5 Answer the following questions.

- a) Write and explain an algorithm for Breath First Search (BFS) traversal for undirected graph with a suitable example. 04
- b) Draw the adjacency matrix of the following graph and find A^2 from it. 04



- c) Construct a binary search tree (using Alphabetic order) for the following data. 04
K, O, X, C, A, B, F, P, U, G, J, Z
Show its Preorder, Inorder and Postorder traversing sequences.

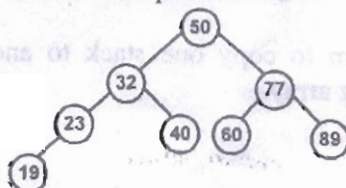
Q.6 Answer the following questions.

- a) Create a max heap tree and sort the given values in ascending order using heap sort with details. 06
4, -12, 16, 32, 24, 57, -11, 69, 96
- b) Calculate total number of comparisons and exchanges required to arrange following data into ascending order using Selection and Bubble sort. 06
99, 56, 12, 16, 98, 34, -22, 10, 0, -10

OR

Q.6 Answer the following questions.

- a) Insert 45, 12, 25 and 39 into the following AVL Tree. Draw balanced tree after each insertion with balance factor and pivot node. 06



- b) Define 2-3 tree, Explain the insertion operation in the 2-3 tree with all cases. 06